PhenoFabric State & Forward Docket

Evidence-bound state and forward path, Snapshot pf-2026-10-17T08:55Z-58176d1

2026-10-17 01:55 PDT · commit 58176d1

Subject: `phenotype-fabric` (`git@github.com:KooshaPari/PhenoFabric.git`) **Snapshot ID:** `pf-2026-10-17T08:55Z-58176d1` **Commit under assessment:** `58176d1` (full SHA in `snapshot.json`) **Observation window:** 2026-10-17T03:18Z – 08:55Z **Generated:** 2026-10-17T08:55Z / 2026-10-17 01:55 PDT **Baseline for deltas:** `pf-2026-10-16T21:51Z-c9eb1a3` (the audit snapshot) **Contract:** State & Forward Docket v1.0 (`handbook/specs/assessment/state-forward-docket-v1/CONTRACT.md`, branch `docs/high-velocity-state-forward-docket-20261016`)

> ***Publication status.*** *This docket is a working-tree/pushed-branch artefact. It is* ***not*** *a release, not installed, not accepted by any consumer, and no human has validated it. Those are separate outcomes and are not claimed.*

0. Two-minute cockpit

The three questions, kept separate

Question	Answer
Implementation — what is built?	22 crates, 53,825 production LOC, 771 unit + 66 integration tests, clippy clean. Transport selection now has a real second adapter; federation is reachable from the wire.
Usability — which beneficiary completes which journey on which artefact?	**One journey, partially.** A developer can clone, build, run the daemon unauthenticated, and get wire/CLI responses. Two-device journeys remain impossible. No GUI journey is proven.
Validation — has a real person used it and did it help?	**NO.** Zero user sessions. Zero feedback. Nothing in this docket is user-validated.

Progress bars

Definitions: leaves are weighted predicates in `snapshot.json`. `P` = accepted pass weight, `F` = fail weight, `U` = unknown/unexecuted/stale weight. **Assessed pass rate = P/(P+F). Assessment coverage = (P+F)/(P+F+U). Verified satisfaction = P/(P+F+U).**

```
GOAL BRANCH                P/(P+F)   COVERAGE   VERIFIED   STATE
A1 single-device usable    [#######-]  8/9      [######--] 6/9   code done; needs a human
A2 two-device usable       [###-----]  4/8      [###-----] 4/8   wire real; cross-device unproven
A3 transport selection     [#####---]  5/9      [#####---] 5/9   executor + 2 adapters; PCIe/RDMA
UNKNOWN
Cross-cutting assurance    [########]  8/8      [########] 8/8   gates green at 58176d1
-------------------------------------------------------------------------
PRODUCT (unique leaves)    [######--] 25/34    [#####---] 23/34
```

`U` is non-zero on A1 (human acceptance unrun), A2 (cross-device unrun), A3 (5 hardware-gated transports). A zero denominator would be undefined, not 100%; none occurs here.

Delta from the audit baseline

Indicator	Baseline `c9eb1a3`	Now `58176d1`	Δ
Unit tests passing	751	**771**	+20
Integration tests passing	63	**66**	+3
`RouteExecutor` exists	**NO**	**YES**	earned
Non-TCP adapter implemented	**NO**	**YES** (`unix_socket`)	earned
Federation reachable from wire	**NO**	**YES**	earned
Silent-no-op BLOCKERs	3	**0**	earned
First-run documentation	absent	`FIRST-RUN.md`	earned
Usable journeys	**0**	**0**	unchanged
Human validation	**0**	**0**	unchanged

Bars: `[####....]` newly earned · `[========]` retained · `[........]` remaining/unmeasured. Nothing was lost or invalidated in this window; one regression was reverted and parked (`wyvern`'s payload work, §7).

What a person can use today

```
cargo run -p fabric-daemon            # unauthenticated local mode, binds 127.0.0.1:9400
cargo run -p fabric-cli -- status     # returns real JSON from a real daemon
```

That is the complete list. Everything else is either unproven or impossible without a second machine, a WorkOS account, or hardware this host does not have.

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1. Intent — the mature product, then its projections

Mature intended product (README.md:9, pinned): a packaged all-in-one computing environment that makes connected machines, OS sessions, VMs, accelerators, displays, audio, input, storage and remote locations appear as **one coherent interactive computer**, while preserving the physical facts needed for good placement decisions.

Closest Viable Product (CVP) — the smallest living slice that preserves that identity: one machine runs the daemon; a second machine connects over a selected non-TCP-by-default transport; a user drives input on one and sees output on the other. Identity-preserving because it exercises placement, transport selection and interactive I/O together.

MVP (narrower, honours the same identity): one machine, daemon + GUI, a user performs one real task and observes a durable result.

Intent archaeology — what the project actually asked for

Source	Authority class	Status
`README.md:6` — "Research and architecture baseline; not an implementation-complete claim"	direct requirement	**accepted, accurate**
`STATUS.md:4` — lifecycle `specified`	direct requirement	**accepted, accurate**
37 ADRs in `docs/adr/`	accepted design	mixed: many partially or wholly unimplemented
A1/A2/A3 acceptance goals	operator-set acceptance	**A1 needs a human; A2/A3 unaccepted**
Release notes R0/R1, `specs/025`, `DEVELOPMENT_GUIDE.md`	historical claim	**contradicted by code** (§8)

Unresolved contradiction (recorded, not resolved): the README and `STATUS.md` describe a specification baseline, while release notes describe delivered capability. 55% of sampled documentary claims are wrong or stale. This docket is the current status layer; the release notes are not.

Superseded intent: none identified in this window.

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2. Usability — the named-beneficiary audit

#	Beneficiary	Journey	Artefact	Environment	Can they complete it?	Evidence
U1	developer	clone → build → daemon responds	`fabric-daemon`	macOS, local	**YES**	unauthenticated mode works
U2	developer	query daemon via CLI	`fabric-cli`	local	**YES**	11 subcommands dispatch
U3	developer	first-run as documented	`install.sh`	clean machine	**UNKNOWN**	installer repo-identity claim unverified
U4	developer	single-device real task	daemon + GUI	local	**NO**	GUI not built by `cargo build --workspace`; zero GUI tests
U5	developer	two-machine topology sync	two daemons	two hosts	**NO**	wire arm exists; no pairing procedure shipped
U6	developer	drive input on A, see output on B	frame plane	two hosts	**NO**	input ownership enforced; no cross-host path
U7	operator	authenticated session	WorkOS	browser	**NO**	needs a human sign-in (§9)
U8	developer	non-TCP transport actually used	executor	one host	**PARTIAL**	`unix_socket` works; tier still selects TCP for L3/L4/L8

Technical E2E proof ≠ user validation. U1 and U2 are proven by real execution. U5 is proven only as loopback (two processes, one host) — that is the wire arm, not cross-device behaviour. **No journey has been validated by an actual user.** None is claimed.

3. Capability and coverage matrix

Capability	Intended	Implemented	Reachable	Exercised	Useful	Accepted
Wire frame plane (dispatch, ack, cancel, reclaim)	✓	✓	✓	✓	partial	—
Reconnect / resume	✓	✓	✓	✓	partial	—
Per-session input ownership	✓	✓	✓	✓	partial	—
RouteExecutor seam	✓	✓	✓	✓	—	—
`unix_socket` transport	✓	✓	✓	✓	—	—
`tcp_transport`	✓	✓	✓	✓	—	—
`quic_transport`	✓	✗	✗	✗	✗	—
`shm_copy`	✓	✗	✗	✗	✗	—
PCIe / RDMA / vhst / DMA-BUF	✓	✗	✗	✗	✗	—
Federation topology merge	✓	✓	✓	✓	partial	—
Graph engine (leases, failover, multihop)	✓	✓	partial	✓	—	—
Capability layer (signing, trust root)	✓	✓	✓	✓	—	—
Desktop GUI	✓	✓	✗	✗	✗	—
Web frontend	✓	✓	✗	partial	✗	—
Authenticated sessions	✓	✓	✓	✗	✗	—
C ABI (`fabric-capability-ffi`)	✓	✓	?	✗	✗	—

Legend: ✓ proven · ✗ absent/unproven · — not assessed · ? unknown. “Reachable” means callable from a production entry point. “Exercised” means a test or run observed it.

Zero-test crates (4): `fabric-gui` (2,608 LOC of GUI), `fabric-tui`, `phenotype-manifest`, `fabric-capability-ffi` (a C ABI). These are unverified surface; the workspace pass count hides them.

4. Unfinished inventory

Scored at `58176d1`. Effort in focused sessions; ranges, not invented hours.

ID	Location	Goal	Present behaviour	User consequence	Closure oracle	Effort
G-01	`multihop/mod.rs:211`	A3	`_ => "tcp_transport"`: L3/L4/L8 silently resolve to TCP	a caller asking for RDMA silently gets TCP	tier either errors or reports the transport actually used	S
G-02	`fabric-gui`	A1	2,608 LOC GUI, **zero tests**	nothing proves it renders live state	headless smoke asserting live daemon state	M
G-03	`install.sh`	A1	repo-identity claim unverified	a new user may 404	verified end-to-end install on a clean machine	S
G-04	two-device pairing	A2	no shipped procedure	two machines cannot be paired by a user	documented + tested pairing flow	M
G-05	`fabric-capability-ffi`	—	C ABI, zero tests	no proof the ABI is correct	ABI-level test	M
G-06	PCIe/RDMA/shm adapters	A3	catalog labels only	locality claim unbacked	adapter + measured delta	**UNKNOWN**
G-07	daemon signal handling	ops	not wired as implied	Ctrl-C behaviour differs from expectation	signal test	S
G-08	graceful shutdown	ops	does not drain connections	in-flight frames can be lost	drain test	S
G-09	concurrent SQLite WAL	data	untested	corruption risk unquantified	two-writer test	S
G-10	wire version mismatch	wire	no test	opaque failure between versions	clear refusal test	S
G-11	capability revocation	security	revocation recorded, not enforced on live handle	a revoked capability may persist	revocation test on open session	M
G-12	docs drift	—	55% of sampled claims wrong/stale	users misled	claims corrected or retracted	M
G-13	release notes R0/R1, spec 025	—	describe undelivered capability	stakeholder misreads progress	annotate as planning records	S
G-14	`deny.yml || true` ×3, `spec-validation.yml:109`	assurance	checks report, cannot fail	a vulnerable dep passes	remove the waivers	S
G-15	payload decoding	A2	**PARKED** (`wyvern`): broke 16 tests, decoder unwired	frame bodies are not decoded	re-integrate AND wire into handlers	M

Closed in this window: G-16 `RouteExecutor` absent → **closed**; G-17 `unix_socket` absent → **closed**; G-18 federation unreachable → **closed**; G-19 three silent no-ops → **closed**; G-20 first-run docs absent → **partially closed** (written, not user-tested).

Not deferred as "post-MVP": every item above is in scope. G-06 is the only genuinely hardware-gated one, and it is marked UNKNOWN rather than deferred.

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5. Assurance

Gate	Method	Observed at `58176d1`
`cargo check --workspace --all-targets`	executed	exit 0
`cargo clippy --workspace --all-targets -- -D warnings`	executed	exit 0, 0 errors
`cargo test --workspace --lib`	executed	**771 passed, 0 failed**
`cargo test -p fabric-integration-tests`	executed	**66 passed, 0 failed**
`cargo fmt --all -- --check`	executed	exit 0
manifest digest	executed	1100 files, all digests verified
CI / Spec Validation / E2E	executed	see `snapshot.json`

Known assurance weaknesses (separate from verdicts): - `cargo test --workspace --lib` skips ~83 main-only unit tests — the reported count understates the suite and overstates coverage. - ~73% of "integration" tests never cross an OS boundary. - Two CI checks are `|| true`: they report and cannot fail. - No negative controls for the hardware-gated transports because they do not exist. - **Applicability ≠ execution ≠ verdict.** A passing fixture qualifies a fixture, not a product.

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6. Forward path

Dependency-ordered. `S` ≈ one focused session.

```
DONE  P0 no-op BLOCKERs        P1 RouteExecutor seam
      P2a unix_socket          P2b federation wire arm
      P2c first-run docs
──────────────────────────────────────────────────────
NEXT  G-01 transport fallback truth        S   no deps   [operator decision]
      G-14 remove CI waivers               S   no deps
      G-07/08/09/10 lifecycle+data tests   2S  no deps
      G-02 GUI headless smoke              2S  no deps
      G-04 two-device pairing procedure    2S  needs P2b ✔
      G-11 revocation enforcement          1S  no deps
      G-12/13 docs truth pass              1S  no deps
      G-15 payload decoding (re-do)        2S  needs wiring
      G-05 C ABI tests                     1S  no deps
LATER G-06 PCIe / RDMA / shm           UNKNOWN   hardware-gated
      CVP completion (2 machines, real E2E)  4–6S + 2 real hosts
```

Time-boxed milestones

Horizon	Deliverable	Depends on
30 min	G-01 decision applied + G-14 waivers removed; CI re-verified	operator answer on G-01
1 hour	+ G-07/08/09/10 lifecycle and data tests green	none
4 hours	+ GUI headless smoke (G-02) + revocation enforcement (G-11) + docs truth pass	none
24 hours	+ two-device pairing procedure (G-04) + C ABI tests + payload decoding redone	a second host for the pair test
CVP	two machines, real transport, real input/output, documented pairing	**4–6 sessions + 2 real machines + operator sign-in**
Mature	+ QUIC/SHM adapters with measured deltas	PCIe/RDMA **UNKNOWN**

Aggregate agent effort vs elapsed time. The 4-hour band assumes parallel lanes on disjoint files, which is how this window ran. Elapsed critical-path time is longer than aggregate effort because CI runs ~4 minutes per push, gates are serial, and the manifest ordering is a strict sequence. Review and rework are not included and have historically been ~30% of lane output (all three lanes this window left work unfinished that I had to complete).

Scenarios: low = 4 sessions to CVP (2 hosts available, no surprises); likely = 6–8 (lane rework, one regression cycle); high = 12+ (hardware for G-06 turns out to be required for acceptance). These are engineering ranges with stated assumptions, **not** statistical P50/P90.

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7. Decisions, blockers, risks

Blockers with precise consequences

Blocker	Precisely prevents	Does not prevent
No WorkOS sign-in by a human	A1 acceptance; any authenticated journey	unauthenticated local work, all wire/graph/transport work
No second physical machine	real cross-device proof (A2 acceptance)	loopback wire-arm proof, which is already green
No RDMA/PCIe hardware on this host	G-06, the A3 upper bound	unix_socket + QUIC + SHM on one Linux host
Operator decision G-01 (A vs B)	closing the silent-degradation finding	the executor seam, which is done

Decisions taken in this window

Decision	Rationale
Revert and park `wyvern`'s payload work	it broke 16 existing tests and its decoder was proven unused by clippy; regressing green for unverifiable gain is not acceptable
Complete the lanes' unfinished work rather than revert it	the underlying work was sound (+20 unit, +3 integration) and the gaps were small and precisely identifiable
Leave `_ => "tcp_transport"` unchanged	the A-vs-B choice is the operator's; inventing intent would bake a guess into the architecture
Report `unix_socket` as proven, quic/shm as absent	a catalog label is not an implementation

Risks

Risk	Severity	Mitigation
Lane work is unfinished more often than reported	high, observed 3/3 this window	independent gating of every lane; never trust a lane's self-report
Manifest ordering is a strict sequence that fails silently	medium, bit twice	codified: `add -A` → regenerate → check → `add MANIFEST` last
Docs drift re-accumulates	medium	this docket set becomes the status layer
GUI is entirely unverified	medium	G-02
Payload decoding is parked, not done	medium	G-15

8. Documentation trust

Document class	Trustworthy as current state?
`README.md` status lines, `STATUS.md`	**yes**
`docs/adr/` (37)	yes as *decisions*; a large fraction unimplemented
`docs/reports/2026-10-17-full-state/`	**yes** — the audit
This docket	**yes** for the instant it describes; superseded on change
Release notes R0/R1, `specs/025`, `DEVELOPMENT_GUIDE.md`	**no**
`docs/runbooks/workos-infisical-shared.md`	**no** — commands fail as written

9. Lineage

Not a fork. PhenoFabric is a first-party project (`KooshaPari/PhenoFabric`, created 2026-09-10). There is no upstream to diverge from and **no fork acceptance criteria apply**. This is recorded explicitly because the Docket contract requires the distinction: applying upstream-parity or migration obligations here would be inventing requirements.

10. Limits of this docket

- No GUI was launched. The GUI is **UNKNOWN**, not broken.
- No second machine was used. Cross-device is **UNPROVEN**.
- No human used any of this. **Zero user validation.**
- Lanes did not run cargo (build contention); all test counts are the coordinator's own executed runs at `58176d1`.
- Hardware-gated transports are **UNKNOWN**, not estimated.
- This is not an exhaustive audit of the portfolio, the local machine, or every historical conversation.